NETWORK TRAFFIC ANALYSIS REPORT

Using Wireshark Packet Analyzer

Capture Interface: wlp3s0
Host IP Address: 10.100.14.248
Capture File: wireshark_wlp3s05U4PO3.pcapng

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1. Introduction

This report presents a structured analysis of network traffic captured on a Linux workstation using the Wireshark packet analyzer. The capture was performed on the wireless interface wlp3s0 and filtered by the host IP address 10.100.14.248. The objective is to examine active protocols, evaluate traffic patterns, and document key observations.

2. Capture Environment

Parameter	Value
Capture Tool	Wireshark Packet Analyzer
Interface	wlp3s0 (Wireless LAN)
Capture File	wireshark_wlp3s05U4PO3.pcapng
Host IP	10.100.14.248
Operating System	Linux
Total Packets	~99,000+ (session ended at frame 99,622)
Profile	Default

3. Protocol Analysis

Each protocol was isolated using Wireshark display filters. Screenshots from each filter view are included below.

3.1 General Traffic Overview — TLS & TCP

Filter: ip.addr == 10.100.14.248

The first capture view shows general traffic to and from the host. TLSv1.3 dominates communication with Cloudflare CDN (172.64.155.209) on port 443. TCP ACK frames and TLSv1.2 traffic to 104.18.39.21 are also visible. A total of 6,211 packets (23.9%) matched this filter.

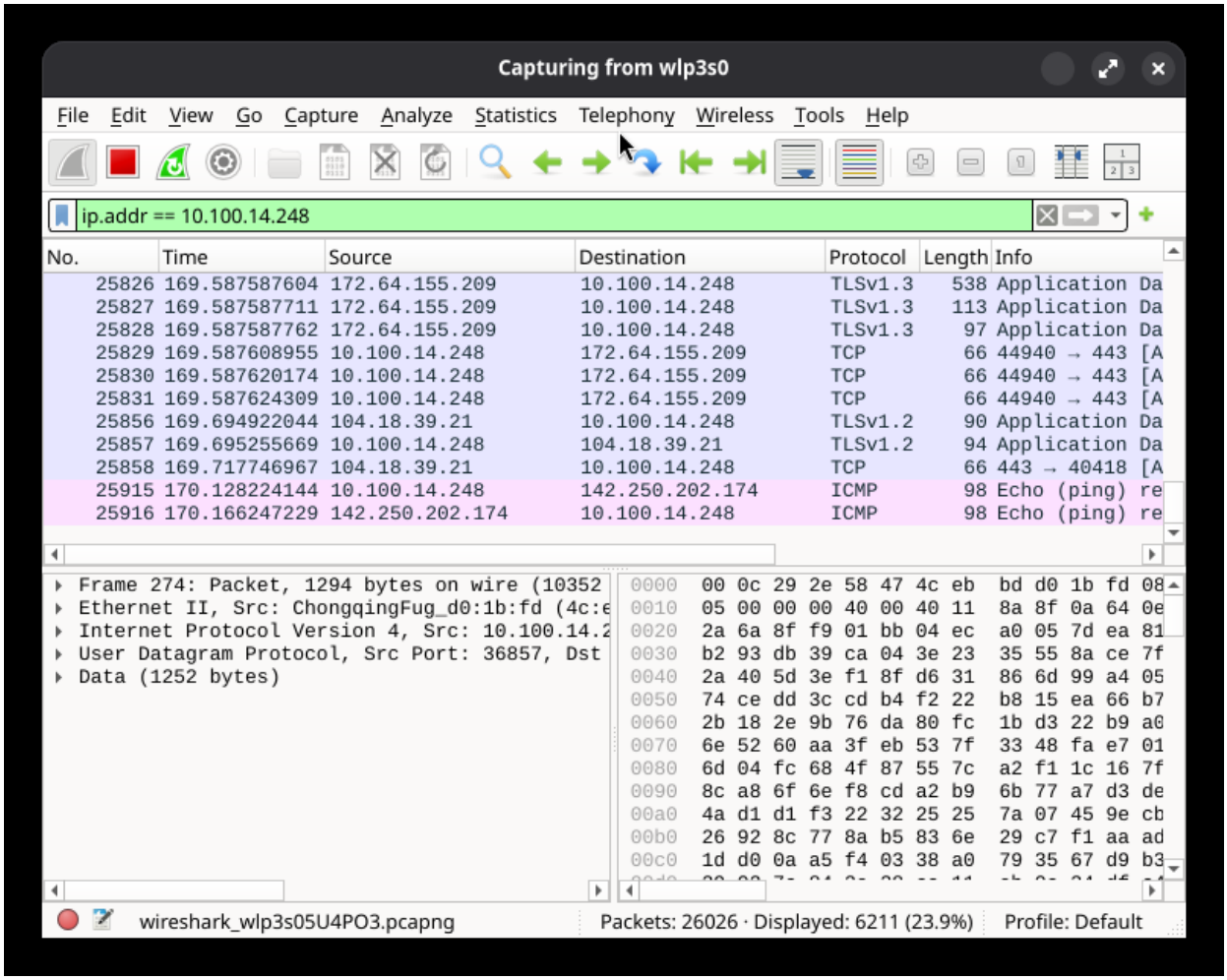

Figure 1: General traffic capture filtered by host IP — showing TLS and TCP sessions

3.2 ICMP — Echo (Ping) Traffic

Filter: ip.addr == 10.100.14.248 and icmp

ICMP Echo Request and Reply packets were captured between the host (10.100.14.248) and Google's server (142.250.202.174). Sequence numbers ranged from seq=84 to seq=92, with 98-byte packets and identifier 0x0719. A total of 176 ICMP packets were displayed (0.3% of total).

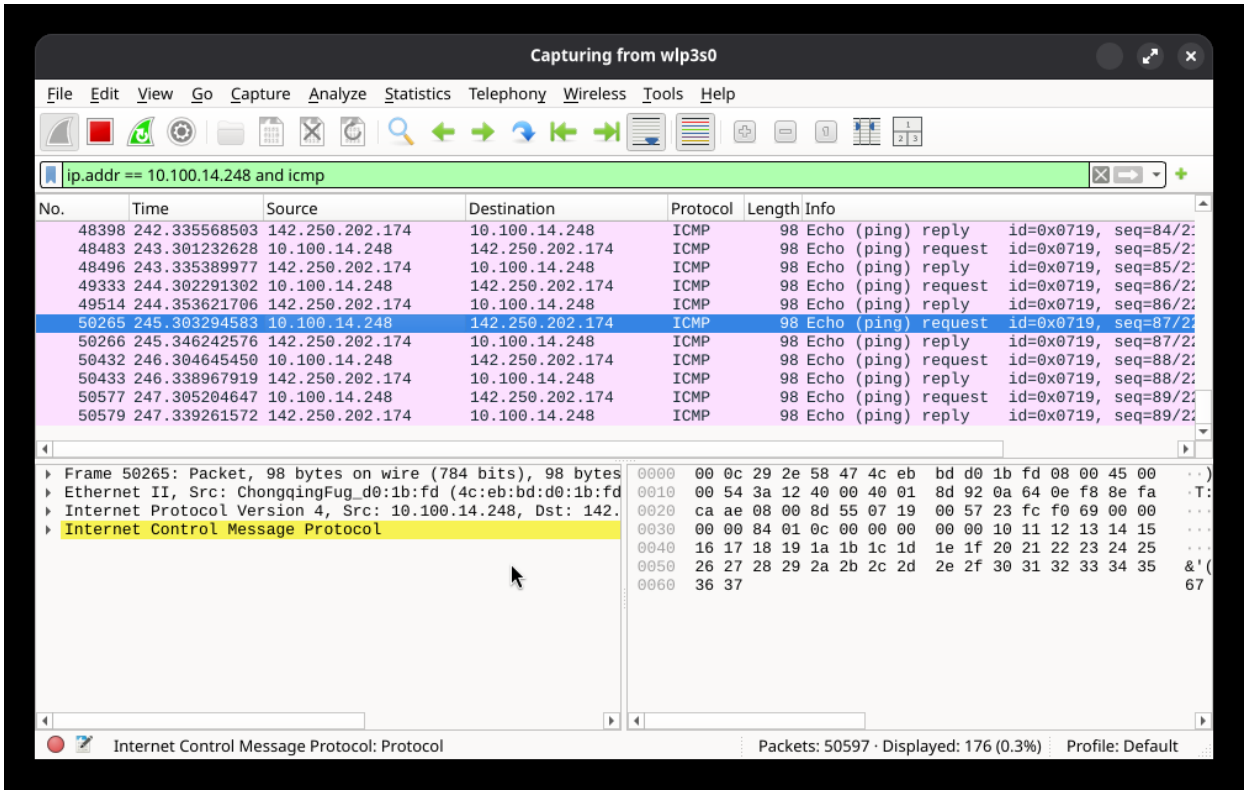

Figure 2: ICMP ping traffic between host and 142.250.202.174 (Google)

3.3 ICMP Packet Detail — Frame 53281

A detailed inspection of ICMP frame 53281 reveals the full packet structure. The source IP is 10.100.14.248 and destination is 142.250.202.174. TTL is 64, ICMP Type is Echo Request (8), checksum is valid (0xc01f), and 40 bytes of ICMP data are present. The timestamp recorded is April 28, 2026.

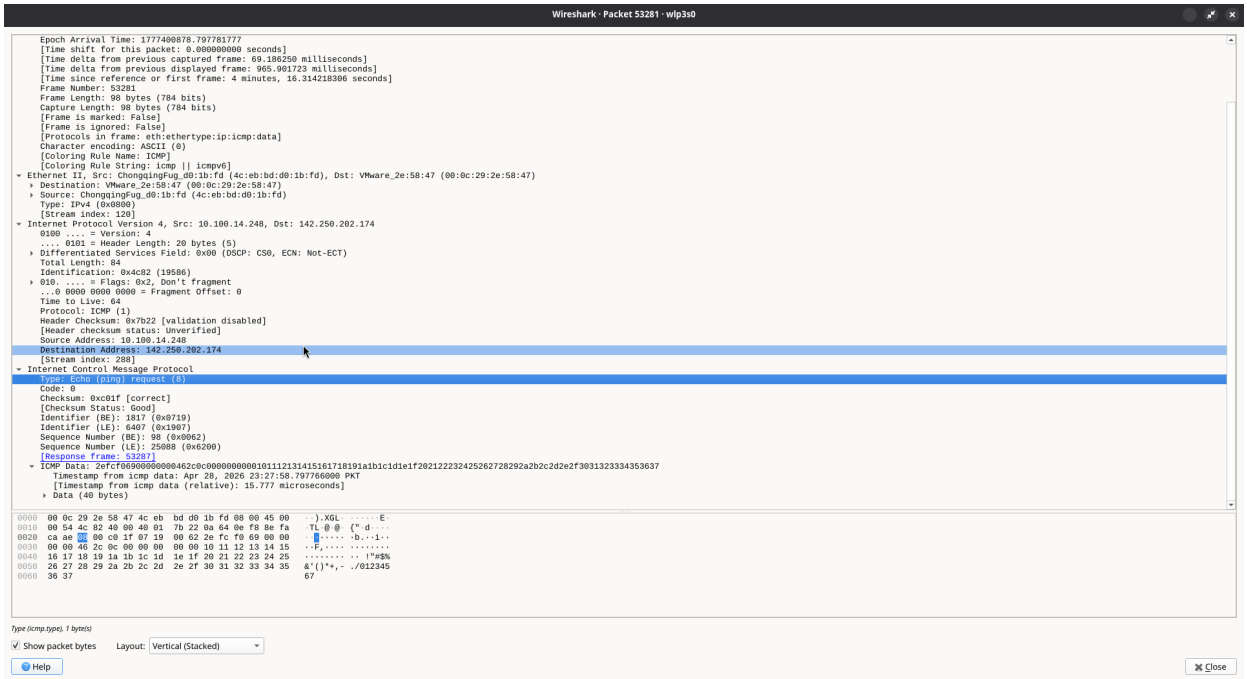

Figure 3: Detailed ICMP packet inspection — Frame 53281

3.4 DNS — Domain Name System

Filter: ip.addr == 10.100.14.248 and dns

DNS queries were directed to Cloudflare's resolver at 1.1.1.1. Both A (IPv4) and AAAA (IPv6) record queries were observed. Multiple NXDOMAIN (No such name) responses were returned for AHSAN-NIAZ.local, indicating failed local hostname resolution. A successful resolution was found for a Google/YouTube streaming endpoint. Total: 163 packets (0.2% of 81,563).

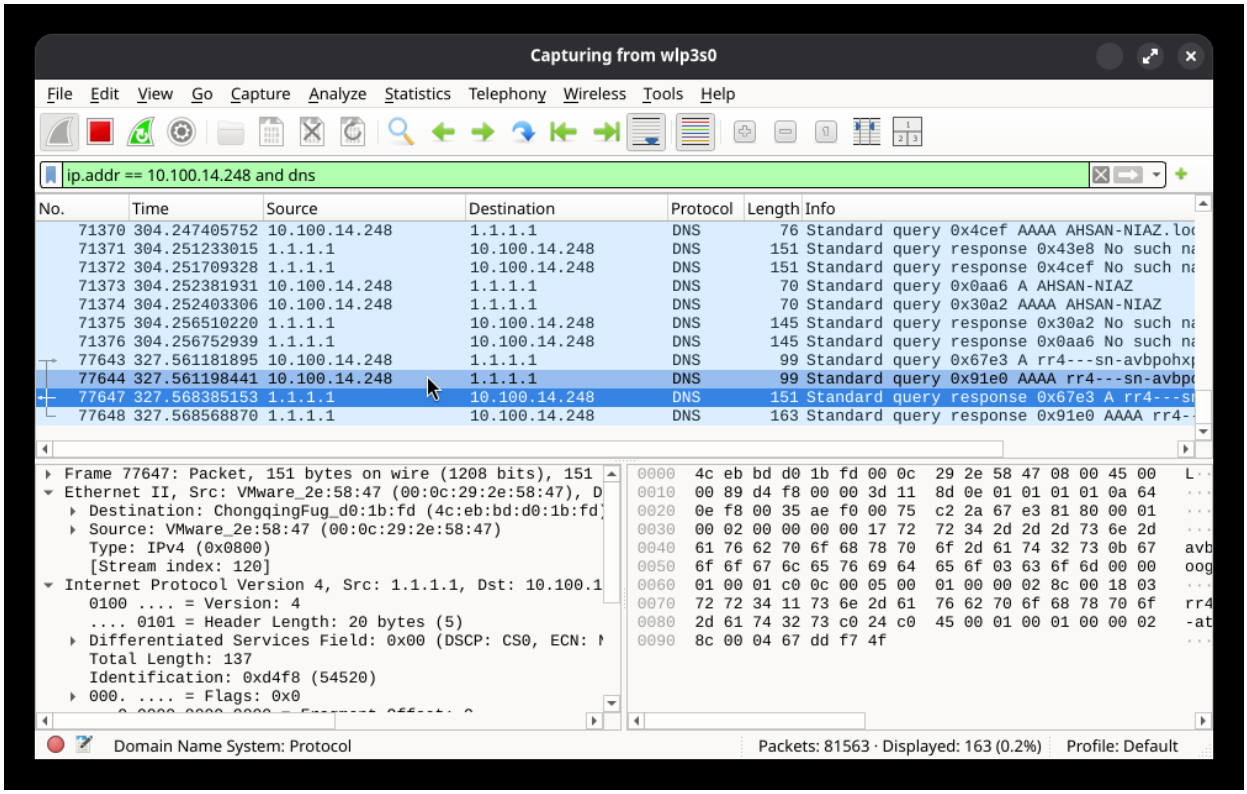

Figure 4: DNS queries and responses filtered by host IP

3.5 TCP — Transmission Control Protocol

Filter: ip.addr == 10.100.14.248 and tcp

TCP traffic consisted primarily of ACK frames maintaining established HTTPS sessions with Cloudflare (172.64.155.209) and Cloudflare CDN (104.18.39.21) on port 443. Packet sizes ranged from 66 bytes (pure ACKs) to 1,492 bytes (TLS Application Data). No retransmissions or RST packets were observed. Total: 3,740 packets (4.3% of 87,129).

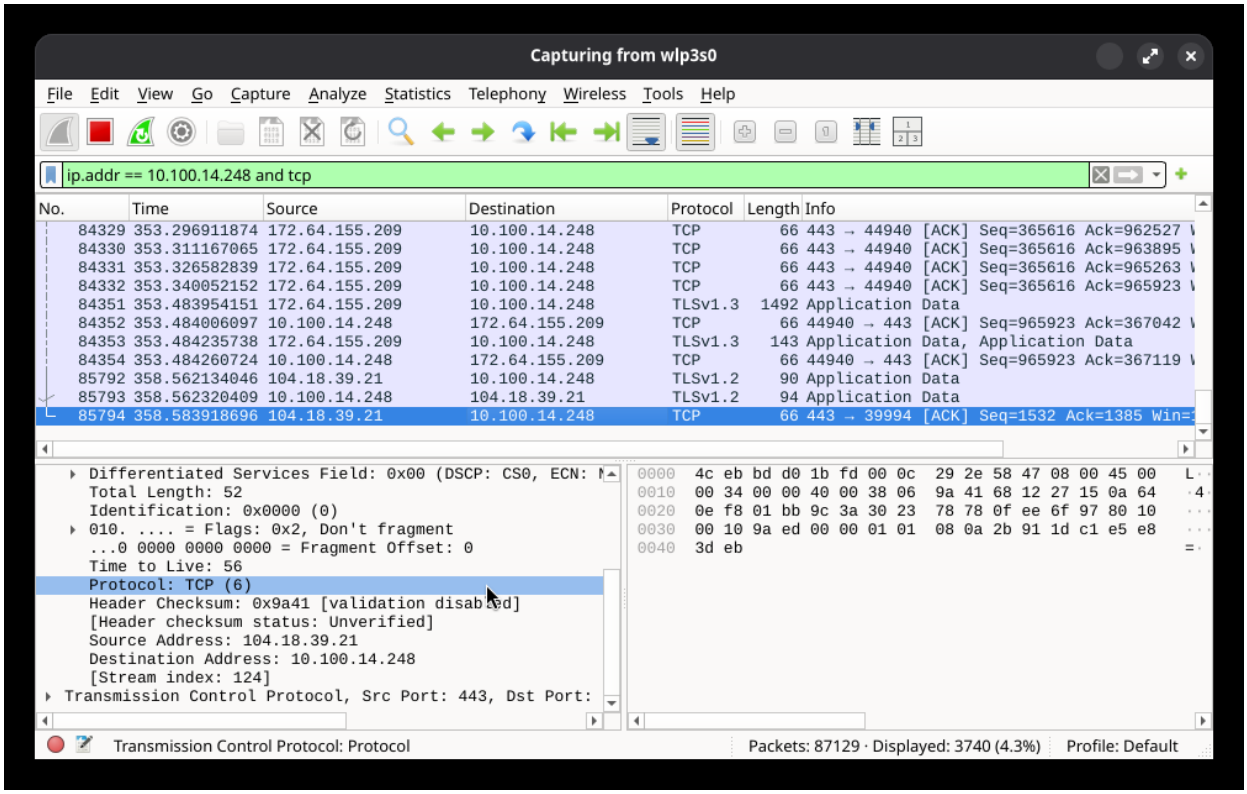

Figure 5: TCP traffic showing ACK and TLS Application Data frames

3.6 UDP / QUIC Protocol

Filter: ip.addr == 10.100.14.248 and udp

UDP traffic was the largest protocol segment at 50.6% of displayed packets. The traffic consists almost entirely of QUIC (HTTP/3) Protected Payload packets exchanged with 103.221.247.79. Two distinct Connection IDs (DCIDs) were observed. Packet sizes ranged from 69 to 1,294 bytes. Total: 45,584 displayed out of 90,167.

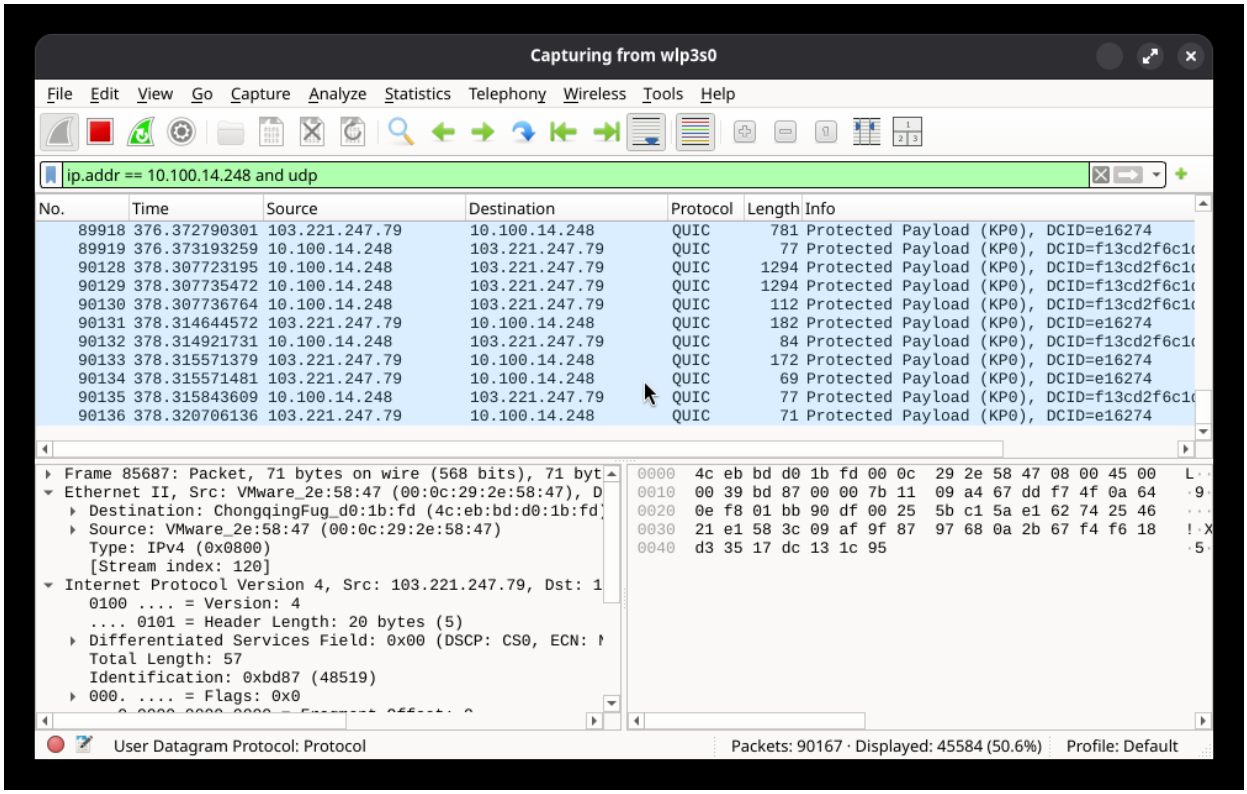

Figure 6: UDP/QUIC traffic — encrypted HTTP/3 payloads to 103.221.247.79

3.7 TLS — Transport Layer Security (Filtered)

Filter: ip.addr == 10.100.14.248 and tls

Dedicated TLS filtering revealed communication with 142.250.202.46 and 104.18.39.21 using TLSv1.2. Application Data frames were exchanged bidirectionally. Fragment Offset was 0 in all frames, confirming no IP fragmentation. Total: 1,404 packets (1.5% of 95,872).

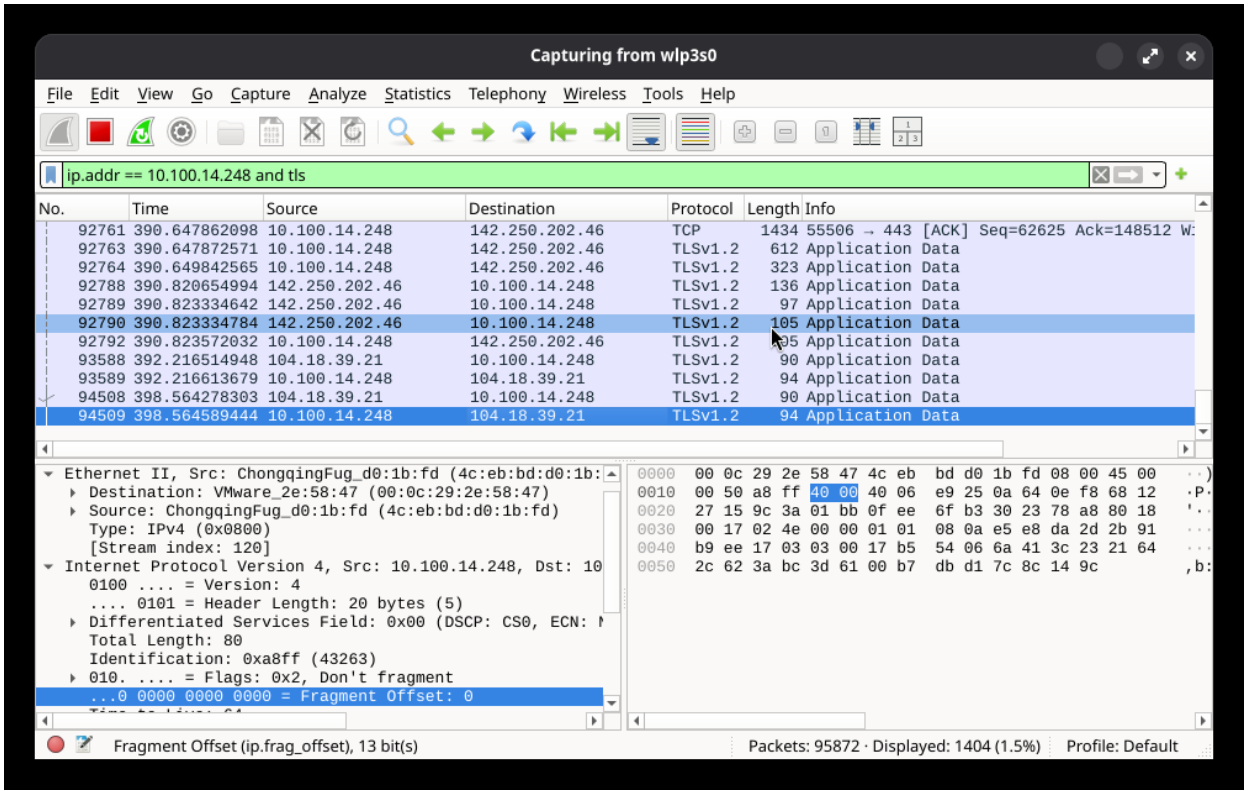

Figure 7: TLS-filtered traffic showing Application Data exchanges

3.8 HTTP — Hypertext Transfer Protocol

Filter: ip.addr == 10.100.14.248 and http

HTTP (unencrypted) traffic was minimal — only 11 packets out of 99,623 total (0.011%). All HTTP traffic was confined to the internal subnet, with POST requests sent to 10.100.5.156, 10.100.26.197, and 10.100.13.1. HTTP 200 OK responses confirmed successful transactions. Large payload sizes (up to 2,424 bytes) indicate REST API calls using UUID-based endpoints.

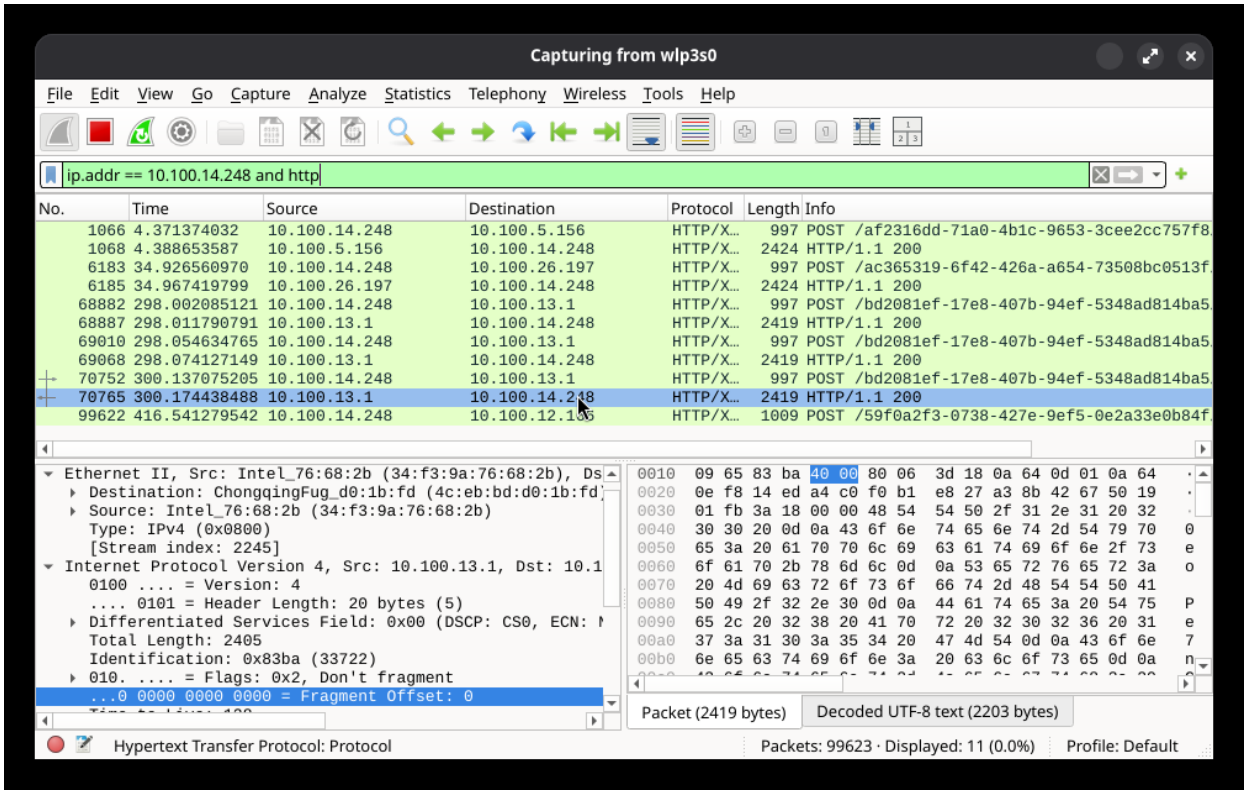

Figure 8: HTTP traffic — internal POST requests and 200 OK responses

4. Traffic Summary

Protocol	Filter	Displayed	% of Total	Key Hosts
TLS (General)	ip.addr == ...	6,211	23.9%	172.64.155.209, 104.18.39.21
ICMP	... and icmp	176	0.3%	142.250.202.174 (Google)
DNS	... and dns	163	0.2%	1.1.1.1 (Cloudflare DNS)
TCP	... and tcp	3,740	4.3%	172.64.155.209, 104.18.39.21
UDP / QUIC	... and udp	45,584	50.6%	103.221.247.79
TLS (Filtered)	... and tls	1,404	1.5%	142.250.202.46, 104.18.39.21
HTTP	... and http	11	0.011%	10.100.5.156, 10.100.13.1

5. Key Findings

- QUIC/UDP traffic was the dominant protocol (50.6%), confirming modern browser (HTTP/3) activity.
- All external communications were encrypted — TLSv1.2 and TLSv1.3 were both in use.
- Successful ICMP ping sessions were conducted to Google servers (142.250.202.174).
- DNS resolution used Cloudflare's 1.1.1.1 resolver; multiple NXDOMAIN failures for AHSAN-NIAZ.local were observed.
- HTTP traffic was strictly internal (local subnet), with no unencrypted external traffic detected.
- No suspicious activity — no port scans, RST floods, or unusual protocol anomalies were identified.

6. Conclusion

The Wireshark capture on interface wlp3s0 reflects typical workstation network behavior. Traffic was dominated by encrypted protocols (QUIC, TLS), consistent with active web browsing. DNS was handled by Cloudflare's 1.1.1.1 resolver. Internal HTTP API activity was observed on the local subnet. No security threats or anomalies were identified within the scope of this capture.

7. Recommendations

- Migrate internal HTTP API endpoints to HTTPS to encrypt all traffic end-to-end.
- Investigate the repeated NXDOMAIN failures for AHSAN-NIAZ.local — a misconfigured hostname may be causing unnecessary DNS lookup overhead.
- Use Wireshark's Protocol Hierarchy Statistics for future captures to get an automated breakdown of protocol distribution.
- Store the .pcapng file securely as it may contain sensitive session metadata.